Lecture 14

Degree, Poincaré–Hopf and de Rham's theorem

Objectives. *We prove that degree is an integer invariant under homotopy for smooth maps between nonempty closed connected oriented manifolds of the same positive dimension. Degree gives the smooth even-sphere hairy-ball and fixed-point theorems. Two cited theorems then relate vector-field zeros and periods to the homology of Part I.*

By the end of this lecture you will be able to:

- Define degree with its dimension, orientation and compactness hypotheses.
- Compute degree as a signed count over a regular value.
- Prove the smooth hairy-ball theorem on even spheres and the smooth Brouwer theorem.
- Compute the two local indices of the specified field on S^2.

Unit 14A Degree

14.1 Integration and top cohomology

Lecture 3 assigned a winding number to a loop in the circle. Give S^1 its counter-clockwise orientation and normalized angular form $\nu = \alpha_{\mathrm{ang}}/(2\pi)$ from Lecture 10. For each $m \in \mathbb{Z}$, the power map and its pullback are

$$f_m \colon S^1 \longrightarrow S^1, \qquad f_m(e^{\mathrm{i}\phi}) = e^{\mathrm{i}m\phi}, \qquad f_m^* \nu = m\nu.$$

Indeed, the angular lift $m\phi$ has derivative m. The integral of the pullback is therefore m. Degree extends this oriented count to higher dimensions. Independence of the chosen integral-one form requires the following fact: a top form on a nonempty closed connected oriented manifold is exact if and only if its integral is zero.

All manifolds below are smooth. Cohomology has real coefficients. On an oriented n-manifold every n-form is closed, since all $(n+1)$-forms vanish. For a closed manifold, Stokes makes integration zero on exact forms. Integration therefore defines a linear map on $H^n(M)$.

Theorem 14.1 (Top cohomology · *sketched*)**.** *For a nonempty closed connected oriented n-manifold M,*

$$I_M \colon H^n(M) \longrightarrow \mathbb{R}, \qquad I_M[\alpha] = \int_M \alpha$$

is an isomorphism. Thus two top forms have equal integrals exactly when they differ by an exact form.

Sketch. For $n = 0$, connected M is a point and integration is its signed value, an isomorphism. For $n \geq 1$, a positive bump function in one positive chart gives a top form of positive integral. Rescaling it gives integral one, proving surjectivity. For injectivity the local input is this compact-support version of the Poincaré lemma: on $\mathbb{R}^n$, a compactly supported top form of integral zero has a compactly supported primitive.

The local construction uses induction and the fundamental theorem of calculus. For $n = 1$, integrate its coefficient from $-\infty$; zero total integral makes the primitive zero also beyond the support. For $n > 1$, write the coefficient as $f(x', u)$, set $h(x') = \int_{\mathbb{R}} f(x', u)\,\mathrm{d}u$, and choose a compactly supported smooth function $\rho(u)$ of integral one. Split the coefficient as $(f - h\rho) + h\rho$. The first term has zero integral in every u fibre, so integrating in u gives a compactly supported primitive, with the sign from moving $\mathrm{d}u$ past $\mathrm{d}x'$. The second term has a primitive by the induction hypothesis applied to $h\mathrm{d}x'$, whose integral is $\int f = 0$, and then wedged with $\rho\mathrm{d}u$.

On M, choose a finite cover by positive coordinate balls diffeomorphic to $\mathbb{R}^n$. A partition of unity splits a top form into ball-supported pieces. Connectedness permits ordering the balls so that each after the first meets an earlier ball. In each chosen overlap take a top form of integral one. Starting with the last piece, subtract its integral times this overlap form and add the same form to the earlier piece. Repeating backwards makes every piece have integral zero, since the total is zero. The local construction now supplies primitives which extend by zero and sum to a global primitive. The cover and transfer details are the sketched steps. See Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 17.30–17.31, pp. 454–455. ∎

For example, on outward S^2, the form $\beta/(4\pi)$ from Lecture 13 has integral one. A 2-form α of integral $a \in \mathbb{R}$ consequently satisfies $[\alpha] = a[\beta/(4\pi)]$.

14.2 The degree of a map

Definition 14.2 (Degree)**.** Let M and N be nonempty closed connected oriented manifolds of the same dimension $n \geq 1$. Let $f \colon M \to N$ be smooth. Choose a top form ν on N with integral one. The *degree* of f is

$$\deg f = \int_M f^*\nu.$$

Degree compares the orientations and will count inverse images with signs.

Theorem 14.3 (Degree and homotopy · *proved given top cohomology and cited smoothing*)**.** *The degree is independent of ν. For every top form α on N,*

$$\int_M f^*\alpha = (\deg f) \int_N \alpha.$$

Homotopic smooth maps have the same degree. The identity has degree one, and a constant

map has degree zero when $n \geq 1$.

Proof. Two choices of ν differ by $\mathrm{d}\eta$, by the top-cohomology theorem. Their pullbacks differ by $\mathrm{d}f^*\eta$, whose integral is zero by Stokes. For an arbitrary α, write $a = \int_N \alpha$. The form $\alpha - a\nu$ has integral zero, so it is exact. Pulling back and integrating gives the claimed formula.

For smoothly homotopic maps, the homotopy operator of Proposition 13.9 makes the pullbacks of ν differ by an exact form. Their integrals coincide. A continuous homotopy between smooth maps can be smoothed by Lee's Theorem 6.29, recalled in Lecture 13, so the same conclusion holds. The identity pulls ν back to itself. A constant map has zero differential, hence pulls a positive-degree form back to zero. ∎

For example, the reflection $(x, y) \mapsto (x, -y)$ on S^1 reverses the angular form, so its degree is -1. The degree also multiplies under composition. For a smooth map g from N to another such manifold of the same dimension, choose a top form ν of integral one on its target. Apply the integral formula to $g^*\nu$ and then to ν to obtain $\deg(g \circ f) = (\deg g)(\deg f)$.

Closedness matters. The map $x \mapsto x/2$ from $[0, 1]$ to itself pulls $\mathrm{d}x$ back to $\mathrm{d}x/2$. The quotient of the two integrals is $1/2$, but the interval has boundary, so this quotient is not a degree under our definition.

14.3 Integrality and regular values

Definition 14.4 (Regular value)**.** A point $y \in N$ is a *regular value* of f if $\mathrm{d}f_x \colon T_xM \to T_yN$ is surjective for every $x \in f^{-1}(y)$. An empty inverse image satisfies this condition.

In equal dimensions each such differential is invertible.

Definition 14.5 (Local degree)**.** The *local degree* at $x \in f^{-1}(y)$, for a regular value y, is $+1$ if $\mathrm{d}f_x$ preserves orientation and -1 if it reverses orientation. Write this sign as $\operatorname{sgn} \det \mathrm{d}f_x$, using positive bases in both tangent spaces.

An inverse branch is a neighbourhood mapped diffeomorphically onto a small target neighbourhood W. The diagram names three such branches V_1, V_2 and V_3 and shows their signs. The proof below constructs the common neighbourhood W.

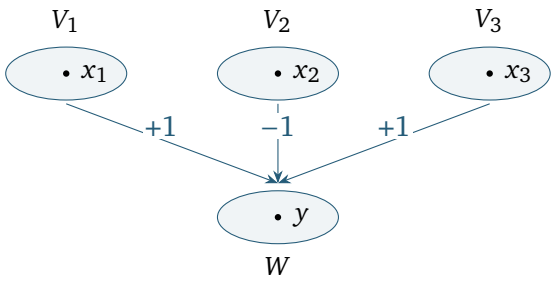


Figure 14.1: Three inverse branches over a small neighbourhood W contribute signs $+1$, -1 and $+1$. Their total contribution is one.

Proposition 14.6 (Signed count · *proved given top cohomology; regular values exist by cited Sard*)**.** *For every regular value $y \in N$, the inverse image is finite and*

$$\deg f = \sum_{x \in f^{-1}(y)} \operatorname{sgn} \det \mathrm{d}f_x.$$

In particular the degree is an integer, independent of the chosen regular value.

Proof. The inverse function theorem isolates each inverse-image point. The fibre is closed in compact M, hence compact. Its isolating neighbourhoods give an open cover with a finite subcover, so the fibre is finite. Write its points as $x_1, \dots, x_m$ when it is nonempty. Choose disjoint neighbourhoods U_i on which f is a diffeomorphism onto a neighbourhood of y. The complement $K = M \setminus \bigcup_i U_i$ is compact and has no point mapping to y. Its image is compact and closed. Choose a small connected coordinate ball W about y inside every $f(U_i)$ and disjoint from $f(K)$. The sets $V_i = U_i \cap f^{-1}(W)$ are then all the inverse branches, each diffeomorphic to W. On a connected branch the orientation sign is constant, since its nonzero Jacobian determinant is continuous.

Choose a smooth top form ν supported in W with integral one. Change of variables on each branch gives

$$\int_M f^*\nu = \sum_{i=1}^{m} \int_{V_i} f^*\nu = \sum_{i=1}^{m} \operatorname{sgn} \det \mathrm{d}f_{x_i}.$$

If the fibre is empty, choose W disjoint from the compact image $f(M)$. The pullback of such a supported ν is zero, giving degree zero and the empty sum. This proves the formula for every regular value.

Sard's theorem says that the critical values of a smooth map have measure zero in the target. We cite Lee, Theorem 6.10, p. 129. Here a critical value is the image of a point with nonsurjective differential. A nonempty coordinate ball has positive measure, so regular values exist. The formula at any of them proves integrality. Since it computes the already defined degree, it also proves independence of the regular value. ∎

For a circle map, the local signs are the signs of the derivative of an angular lift. Consider the lift

$$\psi(\phi) = \phi + \frac{\pi}{2} \sin \phi, \qquad \psi(\phi + 2\pi) = \psi(\phi) + 2\pi.$$

It defines the smooth map $e^{\mathrm{i}\phi} \mapsto e^{\mathrm{i}\psi(\phi)}$. The target point -1 has inverse images at angles $\pi/2$, π and $3\pi/2$. The derivatives there are 1, $1 - \pi/2$ and 1. Their signs sum to one, which is the degree. The graph shows these crossings and a critical level c at a local maximum.

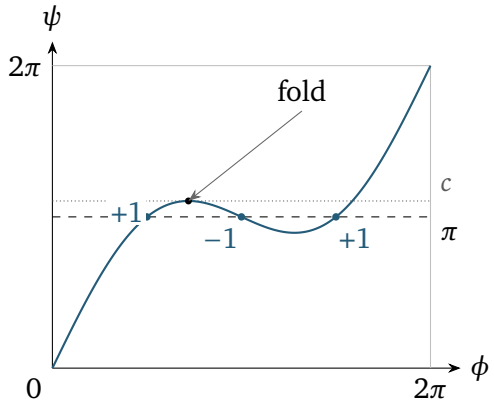


Figure 14.2: The lift crosses level π with signs $+, -, +$. At the critical level c, two crossings meet at a fold; passing below c creates a pair with opposite signs.

The critical angle is $\phi_c = \arccos(-2/\pi)$ and $c = \psi(\phi_c)$. The second derivative there is negative, so the critical point is a nondegenerate local maximum. The signed count stays one on either side.

14.4 Antipodal maps, tangent fields and fixed points

A fixed point of a map f means a point x with $f(x) = x$. Let μ be the outward area form on S^n, characterized in Lecture 12 by $\mu_p(v_1, \ldots, v_n) = \det[p\ v_1\ \cdots\ v_n]$.

Corollary 14.7 (Applications of degree · *proved given top cohomology*)**.** *For $n \geq 1$, the antipodal map $a\colon S^n \to S^n$, given by $a(x) = -x$, has degree $(-1)^{n+1}$. Every smooth tangent field on S^{2m}, with $m \geq 1$, has a zero. Every smooth map from the closed unit ball $\overline{B}^d$ to itself has a fixed point, for $d \geq 1$.*

A nonvanishing smooth tangent field X would give the unit field $u(x) = X_x/\|X_x\|$. Tangency makes x and $u(x)$ orthogonal. The figure shows the semicircle determined by these two unit vectors.

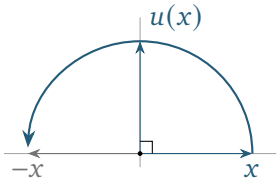


Figure 14.3: In the plane spanned by the orthogonal unit vectors x and $u(x)$, the unit semicircle joins x to $-x$ through $u(x)$.

For the fixed-point argument, the direction from $f(x)$ to x is undefined precisely at a fixed point. The figure displays this direction for $x = (3/4, 1/4)$ and $f(x) = (0, 1/4)$ in the unit disc.

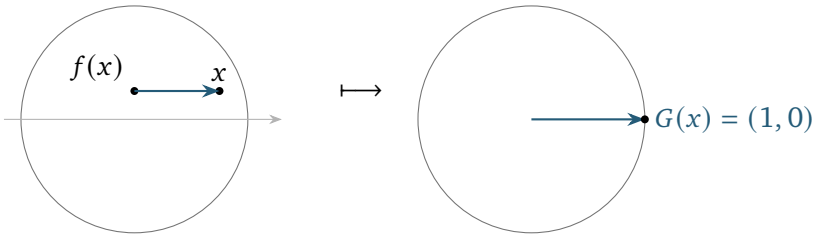


Figure 14.4: The displacement $x - f(x)$ points to the right. Normalizing it gives the sphere-valued map $G(x)$ used under the assumption that no fixed point exists.

Proof. The derivative of the antipodal map sends every tangent vector to its negative. Thus $a^*\mu = (-1)^{n+1}\mu$, because the determinant has $n+1$ columns. The integral definition gives its degree.

Suppose a tangent field X on S^{2m} never vanishes and normalize it to $u(x) = X_x/\|X_x\|$. Tangency gives $x \cdot u(x) = 0$. The map $H\colon [0,1] \times S^{2m} \to S^{2m}$ given by

$$H(t, x) = \cos(\pi t)x + \sin(\pi t)u(x)$$

has squared norm $\cos^2(\pi t) + \sin^2(\pi t) = 1$. It is a smooth homotopy from the identity to the antipodal map. Their degrees would be equal, contradicting $1 \neq -1$. This proves the *hairy-ball theorem* for even spheres.

For the ball of dimension $d \geq 2$, suppose a smooth f has no fixed point. The map $G\colon \overline{B}^d \to S^{d-1}$ given by $G(x) = (x - f(x))/\|x - f(x)\|$ is smooth. Its boundary restriction g

has degree zero: for a normalized top form ν on S^{d-1}, Stokes gives $\int_{S^{d-1}} g^*\nu = \int_{\overline{B}^d} \mathrm{d}G^*\nu = 0$, since $\mathrm{d}\nu = 0$. On the boundary the formula

$$H(t,x) = \frac{x - tf(x)}{\|x - tf(x)\|}$$

is a smooth homotopy from the identity to g. For $t < 1$ its denominator is nonzero because $\|tf(x)\| \le t < 1 = \|x\|$. At $t = 1$ it is nonzero by the assumption. Homotopy invariance gives $\deg g = 1$, a contradiction. For $d = 1$, the function $f(x) - x$ is nonnegative at -1 and nonpositive at 1, so the intermediate value theorem gives a zero. ∎

Evenness is necessary for the tangent-field conclusion. On $S^{2m-1} \subset \mathbb{R}^{2m}$, with $m \ge 1$, the field

$$u(x) = (-x_2, x_1, -x_4, x_3, \ldots, -x_{2m}, x_{2m-1})$$

has $x \cdot u(x) = 0$ because each coordinate pair cancels. Its squared norm is $\sum_{j=1}^{2m} x_j^2 = 1$, so it never vanishes.

The fixed-point proof extends Lecture 3's no-retraction argument: an extension to the ball forces boundary degree zero, whereas a boundary map homotopic to the identity has degree one. The fixed-point conclusion is the *Brouwer theorem*. It also holds for continuous maps. That extension is cited here from Lee, Theorem 17.39, pp. 459–460, for dimensions $d \ge 2$. Its proof uses Theorem 17.38 and smooth approximation into manifolds, Theorem 6.26. For $d = 1$ the intermediate value argument above applies directly to continuous maps. For $d = 2$, Lecture 3 proved it by the retraction argument.

14.5 The index of a vector-field zero

Definition 14.8 (Index of an isolated zero). An isolated zero of a vector field is a zero with a neighbourhood containing no other zero. In dimension $n \ge 2$, choose coordinates centred at such a zero p and a small outward-oriented coordinate sphere S_ε^{n-1}. Write the coordinate components of the field as $X(x) \in \mathbb{R}^n$. Its *index* is

$$\operatorname{ind}_p X = \deg\left(x \longmapsto \frac{X(x)}{\|X(x)\|} : S_\varepsilon^{n-1} \longrightarrow S^{n-1}\right).$$

In dimension one use half the difference of the component's signs at the right and left endpoints of a small coordinate interval centred at the zero. In dimension zero each point has index one.

Changing the small radius gives homotopic maps through spheres avoiding zeros. Independence of the coordinate chart is cited from Milnor, *Topology from the Differentiable Viewpoint*, §6, Lemma 1, pp. 33–35. In dimension two the index is the winding number of the normalized field on a small counter-clockwise circle. For the source (x, y), sink $(-x, -y)$ and centre $(-y, x)$, the target angles are respectively ϕ, $\phi + \pi$ and $\phi + \pi/2$. Each gives index $+1$. For the saddle $(x, -y)$ the target angle is $-\phi$, so the index is -1. The sampling circles and the target dials show these different turns.

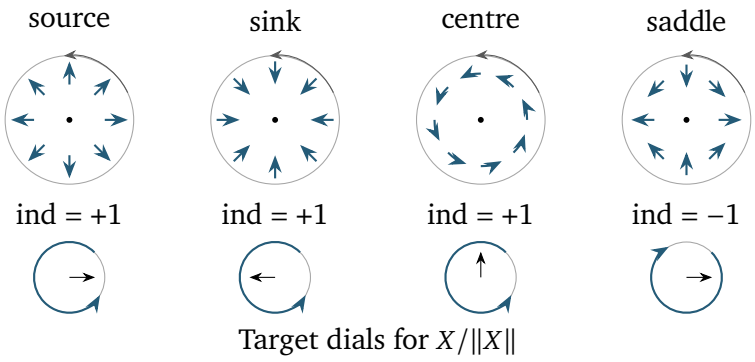


Figure 14.5: Each sampling circle is traversed counter-clockwise. The normalized vectors turn counter-clockwise for the source, sink and centre, and clockwise for the saddle.

In dimension n, the sink $-x$ has index $(-1)^n$ by the antipodal formula on S^{n-1}.

The alternating count of Lecture 6 also has a vector-field interpretation. On a triangulated closed surface, sources at vertices contribute +1, saddles on edges contribute −1, and sinks in faces contribute +1. The triangle illustrates the directions from vertices to edges to faces.

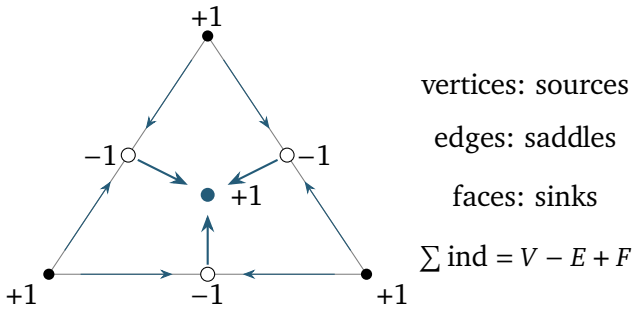


Figure 14.6: One face of a triangulation, with arrows from vertices to edge midpoints and from edge midpoints to the face centre. Counting each vertex, edge and face once gives $V - E + F$.

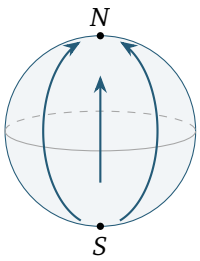
This is an illustration of the index count, rather than a proof that such local pictures fit together smoothly. The cited theorem gives the same total for every smooth field with isolated zeros. For a closed manifold, the Euler characteristic $\chi(M)$ is the alternating sum of its real Betti numbers, as in Lecture 6. The finiteness and the definition for general closed manifolds follow from the comparison in §14.7.

Theorem 14.9 (Poincaré–Hopf · *cited*)**.** *For a smooth vector field with isolated zeros on a closed manifold M,*

$$\sum_{p : X_p = 0} \operatorname{ind}_p X = \chi(M).$$

Here χ is the Euler characteristic. The theorem is cited from Milnor, §6, p. 35.

On the unit sphere write $p = (x, y, z)$ and take $X_p = e_3 - z\,p$. This field provides the explicit test in the figure.



$\operatorname{ind}_N X = +1$

$\operatorname{ind}_S X = +1$

Figure 14.7: The field $X_p = e_3 - zp$ on S^2 points toward the north pole. It vanishes at the north pole N and the south pole S. Each zero has index +1.

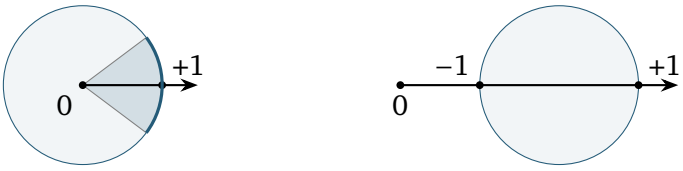
The field is tangent because $p \cdot X_p = z - z\,\|p\|^2 = 0$. A zero requires $e_3 = zp$, hence $p = \pm e_3$. Near either pole use the coordinates $p = (x, y, \sigma\sqrt{1 - x^2 - y^2})$, where $\sigma = +1$ at the north pole and $\sigma = -1$ at the south pole. The coordinate components are $(-zx, -zy)$. On the small circle $(x, y) = \varepsilon(\cos\phi, \sin\phi)$, their normalization is $-\sigma(\cos\phi, \sin\phi)$. Both signs give one counter-clockwise turn as ϕ runs from 0 to 2π. Thus both indices are +1, and their sum is $2 = \chi(S^2)$ from Lecture 6. A sink in two dimensions has index +1 here, not −1. For comparison, the planar source (x, y) has index +1, while the saddle $(x, -y)$ normalizes to $(\cos\phi, -\sin\phi)$ and has index −1. Lecture 9's rotation field L_z has a centre at each pole, with the same index sum. Poincaré–Hopf also rules out a field with no zeros on S^2, since its empty index sum would be zero.

14.6 Radial flux and complex power maps

Let $K \subset \mathbb{R}^3$ be a compact smooth region with connected boundary $S = \partial K$, oriented outward, and assume $0 \notin S$.

Definition 14.10 (Radial Gauss map)**.** The *radial Gauss map* $G\colon S \to S^2$ is given by $G(x) = x/\|x\|$.

It records the direction from the origin. This map differs from the map assigning the outward unit normal to a surface point. The figure shows rays through unit spheres centred at the origin and at $(2, 0, 0)$. A ray from an exterior origin first enters with sign −1 and then exits with sign +1.



$0 \in \operatorname{int} K$ $\qquad$ $0 \notin K$

Figure 14.8: A positive ray exits the sphere centred at the origin once. For the sphere centred at $(2, 0, 0)$ it crosses at $(1, 0, 0)$ and $(3, 0, 0)$ with opposite signs; the shaded sector shows the directions of a surface patch.

For a regular target direction, the inverse images of G are the transverse intersections of its positive ray with S. The sign is positive when the ray exits K and negative when it enters. Indeed, for an outward unit normal n_x and a positive tangent basis (v, w), the sign

of $\det[x\ v\ w]$ is the sign of $x \cdot n_x$. Crossings alternate. Starting inside gives one more exit than entry; starting outside gives equal numbers.

Definition 14.11 (Signed solid angle)**.** The *signed solid angle* of an oriented patch in $\mathbb{R}^3 \setminus \{0\}$ is the integral over that patch of the pullback of the unit sphere's area form under the radial projection $x \mapsto x/\|x\|$.

It measures the area of its radial image with orientation and multiplicity.

Physical input. For an electrostatic point charge $q \in \mathbb{R}$ at the origin, the Coulomb field in rationalised units is given, as in Lecture 12, by

$$\mathbf{E}(x) = \frac{q}{4\pi}\frac{x}{\|x\|^3}, \qquad x \in \mathbb{R}^3 \setminus \{0\}.$$

Its outward flux is the integral of $q\beta_C/(4\pi)$, where β_C is the Coulomb flux form of that lecture.

Example 14.12 (Degree of radial flux)**.** Let β_{S^2} be the outward area form on the target sphere and let $r = \|x\|$. For tangent vectors v and w at $x \in S$,

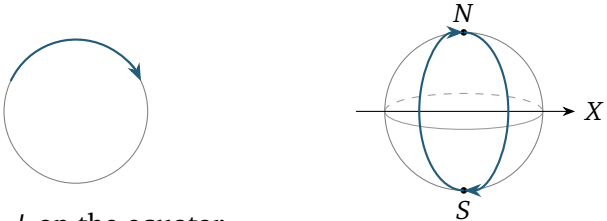
$$\mathrm{d}G_x(v) = \frac{v}{r} - \frac{x(x \cdot v)}{r^3}, \qquad (G^*\beta_{S^2})_x(v, w) = \frac{\det[x\ v\ w]}{r^3}.$$

Indeed, the target form is the determinant with first column x/r. Every additional term parallel to x in the other columns gives zero. This pullback is precisely Lecture 12's Coulomb flux form restricted to S. Since the target area is 4π, its flux calculation gives

$$\deg G = \frac{1}{4\pi}\int_S G^*\beta_{S^2} = \begin{cases} 1, & 0 \in \operatorname{int} K, \\ 0, & 0 \notin K. \end{cases}$$

The flux is therefore an integer multiple of the sphere's area. For the Coulomb field, multiplying this integral by $q/(4\pi)$ gives flux $q \deg G$: it is q when the charge is inside and zero when it is outside.

For complex maps use $\mathbb{C}\mathrm{P}^1$ with its complex orientation: if $z = u + \mathrm{i}v$ is a coordinate, the ordered real coordinates (u, v) are positive. The transition $w = 1/z$ has positive real Jacobian away from zero. This orientation agrees with the outward Bloch sphere of Lecture 7: $z = \tan(\theta/2)e^{\mathrm{i}\phi}$ gives its ordinary angular parametrization. Equivalently, the Bloch map pulls the outward area form back to $4\mathrm{d}u \wedge \mathrm{d}v/(1 + u^2 + v^2)^2$, which is positive. For $z \mapsto 1/z$, the Bloch coordinates change by $(X, Y, Z) \mapsto (X, -Y, -Z)$. This is a rotation by π about the X-axis. The figure compares its equatorial action with its sphere action.



$\phi \mapsto -\phi$ on the equator
degree -1

$(\theta, \phi) \mapsto (\pi - \theta, -\phi)$
degree $+1$

Figure 14.9: The reciprocal map reverses the equatorial direction and interchanges the poles. The two reversals preserve the sphere orientation.

Example 14.13 (Powers on the complex projective line)**.** For $n \in \mathbb{Z} \setminus \{0\}$ consider the map $f_n(z) = z^n$ on $\mathbb{C}\mathrm{P}^1$. For $n > 0$ it fixes zero and infinity. For $n < 0$ it exchanges them. Using $w = 1/z$ near infinity, in the source or target as appropriate, the expression near zero or near infinity is a positive power $w^{|n|}$ or $z^{|n|}$. Thus the extended map is smooth at zero and at infinity.

Fix a finite nonzero target value. Its inverse image has $|n|$ distinct points, by the polar formula for complex roots. At each such point the complex derivative nz^{n-1} is nonzero. Multiplication by a complex number $a + \mathrm{i}b$ has real derivative matrix

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \qquad \det = a^2 + b^2 > 0.$$

Each inverse-image point therefore has local degree +1. The signed-count formula gives $\deg f_n = |n|$. In particular, $z \mapsto 1/z$ has degree +1 on $\mathbb{C}\mathrm{P}^1$. On the equatorial circle the same formula has degree n instead: a negative exponent reverses its one-dimensional orientation. Conjugation $z \mapsto \bar{z}$ has real matrix $\mathrm{diag}(1, -1)$, so it reverses the complex orientation and has sphere degree -1.

Self-check. Identify the degree of a smooth map $f\colon S^1 \to S^1$ with Lecture 3's winding number. In particular, compute it for $f(e^{\mathrm{i}\phi}) = e^{\mathrm{i}n\phi}$.

Answer. Use the normalized angular form $\alpha_{\mathrm{ang}}/(2\pi)$ on the target. A smooth angular lift ψ of the loop $\phi \mapsto f(e^{\mathrm{i}\phi})$ has $\psi(2\pi) - \psi(0) = 2\pi\,\mathrm{wind}(f)$ by Lecture 3. Its pullback is $\psi'(\phi)\mathrm{d}\phi/(2\pi)$, so integration gives $\deg f = \mathrm{wind}(f)$. For the power map the lift is $\psi(\phi) = n\phi$, and the degree is n for every $n \in \mathbb{Z}$.

Self-check. State in one sentence what degree measures and why the integral-one top form was introduced. Include the hypotheses on the source and target.

Answer. For a smooth map between nonempty closed connected oriented manifolds of the same positive dimension, degree is the homotopy-invariant signed count of inverse images at a regular value expressed by integrating the pullback of a top form normalized to integral one. The decisive independence step is that two such forms differ by an exact form, whose pullback integrates to zero by Stokes.

Unit 14B De Rham cohomology and Part I synthesis

14.7 Periods and singular homology

Recall from Lecture 6 that a singular cycle is a chain with zero boundary, and that $H_k(M;\mathbb{R})$ denotes homology with real coefficients. For a smooth singular simplex $\sigma\colon \Delta^k \to M$, integration means $\int_{\Delta^k} \sigma^*\alpha$, extended linearly to chains. A smooth simplex here extends smoothly to a neighbourhood of its parameter simplex. Arbitrary continuous singular simplices cannot be integrated this way.

Theorem 14.14 (De Rham · *cited*)**.** *For every smooth manifold M and $k \geq 0$, integration induces an isomorphism*

$$H^k(M) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(H_k(M;\mathbb{Z}),\mathbb{R}), \qquad [\alpha] \longmapsto \left([c] \longmapsto \int_c \alpha\right).$$

Here $\mathrm{Hom}_{\mathbb{Z}}$ *means homomorphisms of abelian groups. This includes replacing continuous*

cycles by smooth representatives. The cited construction uses Stokes on simplices, including their corners, and shows independence of all representatives. See Lee, Theorem 18.14, p. 484, with the comparison construction in Chapter 18.

A closed form with zero periods over all cycles is exact. Conversely, every homomorphism from integral homology to $\mathbb{R}$ is the period functional of a closed form. Thus the theorem gives both uniqueness of a class from its periods and existence for every such assignment.

Example 14.15 (Periods on the torus)**.** Write $T^2 = \mathbb{R}^2/(2\pi\mathbb{Z})^2$, oriented by the ordered angular coordinates (θ^1, θ^2). The forms $\mathrm{d}\theta^1$ and $\mathrm{d}\theta^2$ descend from $\mathbb{R}^2$ because translations preserve them. Let a and b be the positively oriented coordinate circles. For $\eta_1 = \mathrm{d}\theta^1/(2\pi)$ and $\eta_2 = \mathrm{d}\theta^2/(2\pi)$, the period matrix is

$$\begin{pmatrix} \int_a \eta_1 & \int_b \eta_1 \\ \int_a \eta_2 & \int_b \eta_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

For example, on a the angle θ^1 increases from 0 to 2π and θ^2 is constant. This gives the first column. Lecture 6 gives $H_1(T^2;\mathbb{Z}) \cong \mathbb{Z}^2$, generated by a and b. De Rham's theorem therefore makes $[\eta_1]$ and $[\eta_2]$ a basis of $H^1(T^2)$. The prescribed periods $7/3$ and $-5/2$ are realized by $(7/3)\eta_1 - (5/2)\eta_2$. Neither angle is a global real-valued function on the torus. A nonzero period proves that neither angle form is exact.

The wedge $\eta_1 \wedge \eta_2$ has integral one on the oriented torus, since integration over the angular square gives $(2\pi)^2/(2\pi)^2 = 1$. Together with $H_2(T^2;\mathbb{Z}) \cong \mathbb{Z}$, this gives $H^2(T^2) \cong \mathbb{R}$.

Definition 14.16 (Singular cochains)**.** A real singular k-cochain is a real-linear functional $\varphi\colon C_k(M;\mathbb{R}) \to \mathbb{R}$. Its *coboundary* is the $(k+1)$-cochain given by

$$\varphi \circ \partial_{k+1}\colon C_{k+1}(M;\mathbb{R}) \longrightarrow \mathbb{R}, \qquad c \longmapsto \varphi(\partial_{k+1}c).$$

A cochain is *closed* when its coboundary is zero. A k-cochain is a *coboundary* when it has the form $\psi \circ \partial_k$ for a $(k-1)$-cochain ψ. In degree zero the only coboundary is zero.

The identity $\partial_k\partial_{k+1} = 0$ makes every coboundary closed. For a smooth singular $(k+1)$-chain c and a k-form α, Stokes gives

$$\int_c \mathrm{d}\alpha = \int_{\partial c} \alpha.$$

Thus integration respects the passage from forms to cochains in the cited comparison theorem.

We identify the cohomology of real cochains with the real dual of homology. Write $Z_k = \ker \partial_k$ for cycles and $B_k = \operatorname{im} \partial_{k+1}$ for boundaries. A closed cochain vanishes on B_k, so restriction to Z_k gives a well-defined functional on $Z_k/B_k = H_k(M;\mathbb{R})$. Conversely, composing a functional on this quotient with the quotient map gives a functional on Z_k that vanishes on B_k. Extending a basis of Z_k to a basis of $C_k(M;\mathbb{R})$ extends this functional to a closed cochain. This proves surjectivity of restriction.

For its kernel, let a closed cochain φ vanish on all cycles. For $k \geq 1$, define a functional first on $\operatorname{im} \partial_k$ by

$$\psi(\partial_k c) = \varphi(c), \qquad c \in C_k(M;\mathbb{R}).$$

Two chains with the same boundary differ by a cycle, so this is well defined. Extend ψ to $C_{k-1}(M;\mathbb{R})$ by a basis extension. The defining equality gives $\varphi = \psi \circ \partial_k$, a coboundary. Conversely, a coboundary vanishes on cycles because their boundary is zero. For $k = 0$, every

chain is a cycle because $\partial_0 = 0$, so the kernel consists only of the zero cochain. Restriction therefore identifies closed cochains modulo coboundaries with the real homology dual.

For integer chains, real-valued cochains are additive homomorphisms $C_k(M;\mathbb{Z}) \to \mathbb{R}$. Each extends uniquely to a real-linear functional on $C_k(M;\mathbb{R})$ by its values on singular simplices. The coboundary operations agree. The same restriction and factorization argument for integer chains uses a cited extension property in place of basis extension: every homomorphism from a subgroup of an abelian group to $\mathbb{R}$ extends to the whole group. This property is called injectivity of $\mathbb{R}$ and is stated in Lee, p. 473. It supplies both extensions in the argument above. Thus both homology duals compute the cohomology of the same real-valued cochain complex. This proves their identification given that cited property.

Definition 14.17 (Betti numbers and Euler characteristic)**.** For a closed manifold, its *Betti numbers* and *Euler characteristic* are

$$b_k(M) = \dim_{\mathbb{R}} H_k(M;\mathbb{R}), \qquad \chi(M) = \sum_{k=0}^{\dim M} (-1)^k b_k(M).$$

These dimensions are finite by the cited finite-dimensionality of de Rham groups (Lee, Problem 18-8, p. 489, an exercise with hints) and the comparison just proved. For a finite Δ-complex model, Theorems 6.13 and 6.14 identify this with the count of Lecture 6. In finite dimension the comparison gives $\dim H^k(M) = b_k(M)$. The real coefficients matter: an integer homology class annihilated by $m \geq 1$ has period P with $mP = 0$, hence $P = 0$. Differential forms therefore do not recover torsion classes. For example, Lecture 6's integral homology calculation gives $H_1(\mathbb{R}\mathrm{P}^2;\mathbb{Z}) \cong \mathbb{Z}_2$. Every homomorphism from this group to $\mathbb{R}$ is zero, since its value P on the generator satisfies $2P = 0$. De Rham's theorem gives $H^1(\mathbb{R}\mathrm{P}^2) = 0$, although its fundamental group is nontrivial.

14.8 Loops, homology and periods in Part I

The fundamental group records based loops and their composition. Passing to first homology makes that composition abelian, by the cited comparison in Lecture 6. Higher homology uses cycles of higher dimension. De Rham cohomology describes their real-valued period functionals. The diagram separates the loop comparison from the period comparison. In its last line, M is closed and X is smooth with isolated zeros.

$$\pi_1(M) \xrightarrow{\text{abelianization}} H_1(M;\mathbb{Z})$$

$$H^k(M) \xrightarrow[\text{integration: isomorphism}]{} \mathrm{Hom}_{\mathbb{Z}}(H_k(M;\mathbb{Z}),\mathbb{R})$$

$$\chi(M) = \textstyle\sum_k (-1)^k b_k(M) = \sum_{p:X_p=0} \mathrm{ind}_p X$$

Figure 14.10: For a path-connected space, first integral homology is the abelianization of the fundamental group. For a smooth manifold, de Rham classes correspond to real period functionals in every degree.

The sphere and torus calculations distinguish several roles of these invariants. The groups in each triple below are in degrees 0, 1 and 2; cohomology is de Rham cohomology.

Space	π_1	(H_0, H_1, H_2) over $\mathbb{Z}$	(H^0, H^1, H^2)
S^1	$\mathbb{Z}$	$(\mathbb{Z}, \mathbb{Z}, 0)$	$(\mathbb{R}, \mathbb{R}, 0)$
S^2	$\{1\}$	$(\mathbb{Z}, 0, \mathbb{Z})$	$(\mathbb{R}, 0, \mathbb{R})$
T^2	$\mathbb{Z}^2$	$(\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z})$	$(\mathbb{R}, \mathbb{R}^2, \mathbb{R})$
$\mathbb{R}\mathrm{P}^2$	$\mathbb{Z}_2$	$(\mathbb{Z}, \mathbb{Z}_2, 0)$	$(\mathbb{R}, 0, 0)$
$\mathbb{R}^2 \setminus \{0\}$	$\mathbb{Z}$	$(\mathbb{Z}, \mathbb{Z}, 0)$	$(\mathbb{R}, \mathbb{R}, 0)$

On the circle, a normalized angular period is both winding and degree. On the sphere, the normalized area period computes degree, while the sum of vector-field indices is the Euler characteristic, equal to two. On the torus, two independent angular periods give two classes in degree one. Its Euler characteristic is zero, consistent with a nonzero constant field descending from $\mathbb{R}^2$. The projective plane retains an order-two loop in integral homology, although every closed real 1-form has zero period around it.

Charts and tangent spaces made calculus possible on these spaces. Exterior differentiation and Stokes connected that calculus to cycles. The index of a zero and the degree of a map are integers obtained from this connection. Lecture 15 introduces a metric to measure lengths and define the Hodge star. These measurements add geometric information to the topological invariants above.

Self-check. Consider the real chain complex with $C_2 = \mathbb{R}$, $C_1 = \mathbb{R}^3$ and $C_0 = \mathbb{R}$, whose boundary matrices are

$$\partial_2 = \begin{pmatrix} 1 \\ 5 \\ 1 \end{pmatrix}, \qquad \partial_1 = \begin{pmatrix} 2 & -1 & 3 \end{pmatrix}.$$

For the 1-cochains $\varphi = (7, 1, -12)$ and $\eta = (-8, 4, -12)$, decide which are closed and which are coboundaries. Use the cycle $z = (1, 2, 0)^{\mathsf{T}}$ to distinguish their homology functionals.

Answer. The product $\partial_1 \partial_2 = 2 - 5 + 3 = 0$ verifies the chain condition. Both cochains are closed because $\varphi \partial_2 = 7 + 5 - 12 = 0$ and $\eta \partial_2 = -8 + 20 - 12 = 0$. The vector z is a cycle since $\partial_1 z = 2 - 2 = 0$. The value $\varphi(z) = 7 + 2 = 9$ prevents φ from being a coboundary: every cochain of the form $\psi \circ \partial_1$ vanishes on z. The equality $\eta = -4\partial_1$ exhibits η as a coboundary. Its homology functional is zero.

Summary. On a nonempty closed connected oriented manifold, top cohomology is measured by integration. Between such manifolds of equal positive dimension, degree is a homotopy invariant integer, computed by signed inverse images at a regular value. It gives the even-sphere hairy-ball theorem for smooth fields and the smooth Brouwer theorem. Poincaré–Hopf relates zero indices to the Euler characteristic. De Rham's theorem identifies cohomology with real period functionals on homology. Part II adds a metric, connections and physical fields to these topological constructions.

Reading. Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 17.30–17.31, pp. 454–455; Theorem 17.35, pp. 457–458; Theorems 17.38–17.39, pp. 459–460; Theorem 18.14, p. 484. Milnor, *Topology from the Differentiable Viewpoint*, §6, pp. 32–35 (online scan).